
IAM Users and IAM Groups – Demo Guide

This demo covers:

1. Creating IAM Users
2. Creating IAM Groups
3. Attaching policies to Groups
4. Adding users to Groups
5. Verifying access with permissions

Prerequisites

- AWS account
- IAM Full Access permissions
- Admin access to AWS Console or AWS CLI setup

Step-by-Step: Create IAM Users and Groups

Step 1: Create an IAM Group

 Via AWS Console:

1. Go to **IAM Console** → **Groups**
2. Click “**Create group**”

3. Name the group (e.g., `S3ReadOnlyGroup`)
4. Attach a policy like `AmazonS3ReadOnlyAccess`
5. Click “**Create group**”

👉 **Via CLI:**

```
aws iam create-group --group-name S3ReadOnlyGroup
```

```
aws iam attach-group-policy \  
  --group-name S3ReadOnlyGroup \  
  --policy-arn arn:aws:iam::aws:policy/AmazonS3ReadOnlyAccess
```

✅ **Step 2: Create IAM Users**

👉 **Via AWS Console:**

1. Go to **IAM Console** → **Users**
2. Click “**Add users**”
3. Enter username (e.g., `alice`)
4. Select **Access type**:
 - **Programmatic access** (CLI, SDK)
 - **AWS Management Console access**
5. Click **Next** → **Add to group**
6. Add to the group you created (`S3ReadOnlyGroup`)
7. Click “**Create user**”

👉 **Via CLI:**

```
aws iam create-user --user-name alice
```

```
aws iam add-user-to-group \  
  --user-name alice \  
  --group-name S3ReadOnlyGroup
```

```
--group-name S3ReadOnlyGroup
```

Optional (for console access):

```
aws iam create-login-profile \  
  --user-name alice \  
  --password 'TemporaryPassword123!' \  
  --password-reset-required
```

Optional (for programmatic access):

```
aws iam create-access-key --user-name alice
```

✅ Step 3: Test Permissions

Case 1: User Access Check – Read from S3

If Alice is in **S3ReadOnlyGroup**:

- She can **list buckets** and **view objects**.
- She **cannot upload or delete objects**.

CLI Test (after setting up AWS CLI with Alice's credentials):

```
aws s3 ls  
aws s3 cp s3://your-bucket-name/file.txt .  
aws s3 rm s3://your-bucket-name/file.txt # This will fail (no write access)
```

✅ Step 4: (Optional) Detach or Modify Permissions

To remove access:

```
aws iam detach-group-policy \  
  --group-name S3ReadOnlyGroup \  
  --policy-arn arn:aws:iam::aws:policy/AmazonS3ReadOnlyAccess
```

To add a new policy:

```
aws iam attach-group-policy \  
--group-name S3ReadOnlyGroup \  
--policy-arn arn:aws:iam::aws:policy/AmazonEC2ReadOnlyAccess
```



Summary Notes

IAM Component	Purpose
User	Entity representing a person/service
Group	Collection of users with shared permissions
Policy	Defines what actions are allowed or denied

- Use **Groups** to manage permissions **centrally**.
 - Avoid assigning policies directly to users.
 - Enforce **least privilege** — give users only what they need.
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